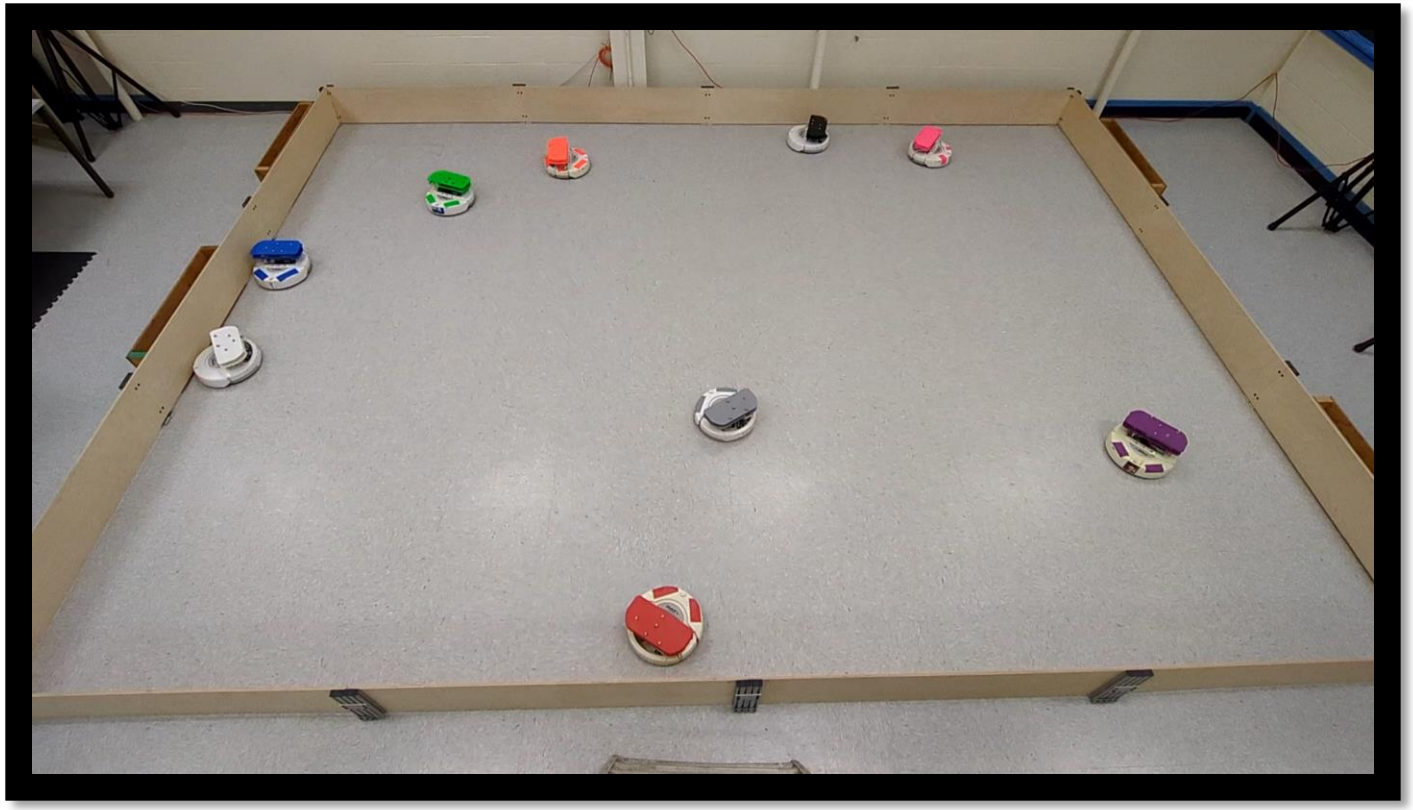
Autonomous Mobile Robots: Final Competition!

MAE 4180/5180, ECE 4180/5772, CS 4758/5758



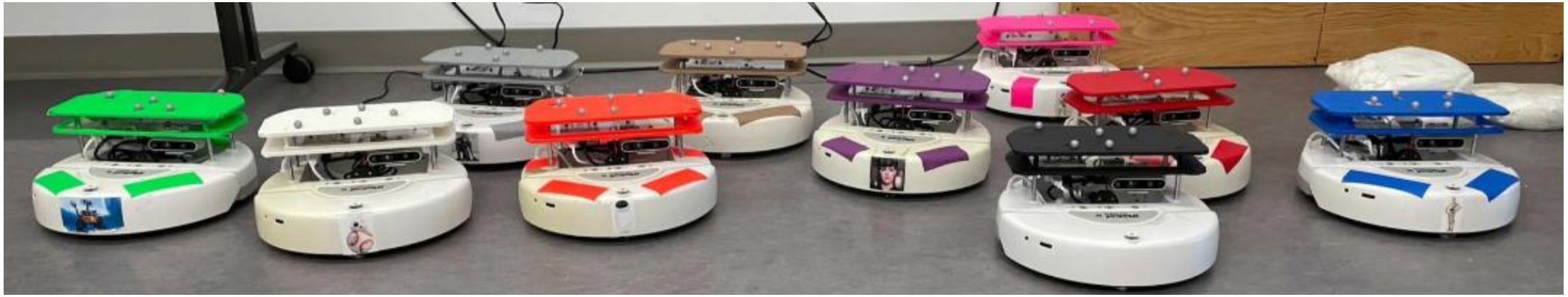
Prof. Anastasia Bizyaeva

Final Competition Spring 2026

Teams of 3 students, **must be enrolled at the same credit level**

Components:

1. Design pitch group presentation (5 minutes)
2. Competition attendance and participation (all team members must be present!)
3. Code (all team members **must** contribute, submitted code must run on competition day)
4. Final report (written **individually** by each team member)



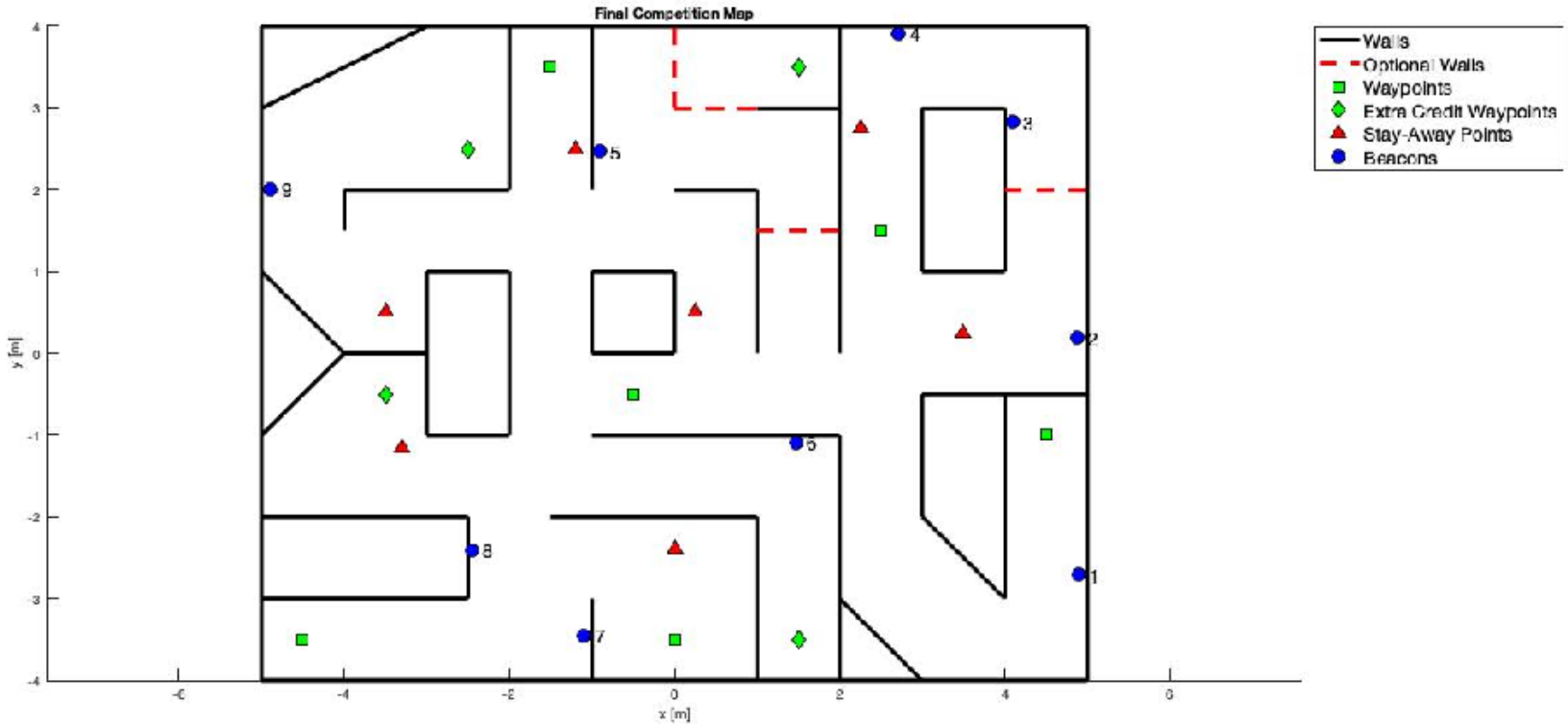
MAE 4180, ECE 4180, CS 4758: **simulator**

Challenge: putting it all together!

1. Localize your robot
   - *No ground truth given!*
   - *Noisy odometry*
2. Produce a "correct" map
   - *Given: map with "optional" walls; figure out which are actually there*
3. Navigate environment
   - *Visit waypoints*
   - *Avoid "stay-away" points*
   - *Indicate when waypoint is reached*

MAE 4180, ECE 4180, CS 4758: **simulator**

Example map (not actual competition map):



MAE 4180, ECE 4180, CS 4758: **simulator**

Practice map will be provided on Canvas; actual competition map revealed day of competition (Thursday April 30).

Scoring breakdown: 5-minute time limit

The team's score for the competition will be calculated as follows:

- 10 points for each correct *waypoint* visited
- 20 points for each correct *ECwaypoint* visited
- -5 points every time the robot indicates incorrectly that it is at a waypoint
- -5 points every time the robot drives over a stay-away point
- 10 points for each optional wall that is correctly determined (is in the workspace or not)
- -10 points for each optional wall that is incorrectly determined (no points are deducted for optional walls that have not been determined)
- 10*(time limit (minutes) - actual time(minutes)) if all *waypoints* and *ECwaypoints* are visited before time runs out

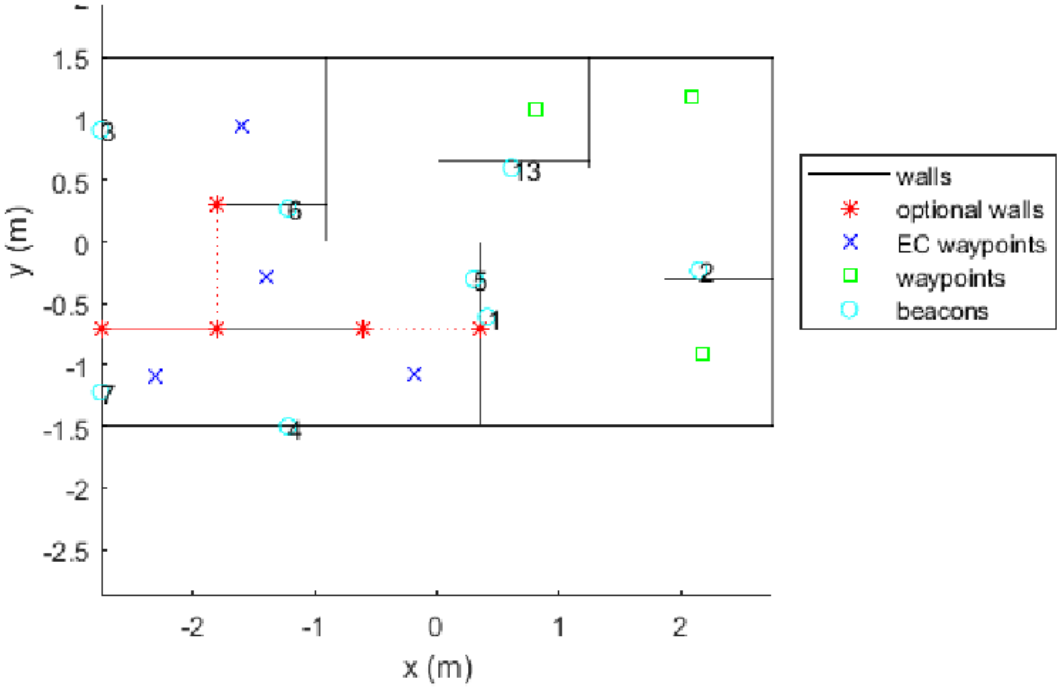
MAE 5180, ECE 5772, CS 5758: **hardware**

Challenge: putting it all together!

1. Localize your robot
   - *No ground truth given!*
2. Produce a "correct" map
   - *Given: map with "optional" walls; figure out which are actually there*
3. Navigate environment
   - *Visit waypoints*
   - *Indicate when waypoint is reached*

MAE 5180, ECE 5772, CS 5758: **hardware**

Example map (not actual competition map):



MAE 5180, ECE 5772, CS 5758: **hardware**

Practice simulator map will be provided on Canvas and a physical map in **open lab hours**; actual competition map revealed day of competition (Tuesday May 5).

Scoring breakdown: 7-minute time limit

The team's score for the competition will be calculated as follows:

- 10 points for each correct *waypoint* visited
- 20 points for each correct *ECwaypoint* visited
- -5 points every time the robot indicates incorrectly that it is at a waypoint
- 10 points for each optional wall that is correctly determined (is in the workspace or not)
- -10 points for each optional wall that is incorrectly determined (no points are deducted for optional walls that have not been determined)
- 10*(time limit (minutes) - actual time(minutes)) if all *waypoints* and *ECwaypoints* are visited before time runs out

Competition logistics: key dates

- Form a team **this week**, submit google form by 11:59 PM Thurs 4/9
  - Teams of 3; A small number of 2-person teams may be permitted
  - All group members must be enrolled in the same number of credits
- **Next week**: design pitch presentations during lecture 4/14 and 4/16
  - Presentation slot sign up google form posted Thursday 4/9; 5 minutes maximum
- 4/24-5/4: open lab hours; 4-credit student teams can sign up for slots to book a workstation; exact hours and signups TBA
- Thursday 4/30: simulator competition (during lecture)
- Tuesday 5/5: hardware competition (during lecture, in 101a Tang lab space)
- 11:59 Tuesday 5/12: final reports due – **NO LATE REPORTS ACCEPTED**

Competition logistics

- Design pitch (40pt): all team members must be present, no makeups
  - Overall strategy, algorithmic choices, work distribution, integration plan
  - Rubric provided on Canvas

- Competition day:
  - Code uploaded to Canvas by 8am; point penalty if it does not run on course staff computer
  - May restart code as many times as needed within the 5/7 min time limit

- Final report (200pt):
  - Follow instructions closely; strict page limits. Be succinct.
  - Rubric provided on Canvas

- Open lab hours (4-credit only):
  - Collect lots of data!